

Поправка задания.

III.1.

I

$$|A| = 10 \quad |B| = 15 \quad |C| = 12$$

$$|U| = 31 \quad \text{ММ}$$

$$|M| = |B \cap C| = 24$$

$$|B \cup C| = |U \setminus A| = |U| - |A| = 31 - 10 = 21$$

$$|B \cup C| = |B| + |C| - |B \cap C|, \Rightarrow |B \cap C| = |B| + |C| - |B \cup C| = 15 + 12 - 21 = 6$$

III.2.

	ЯЗ	ЯЖ	ГЗ	ГЖ	Σ	
Б	$x=28$	?	$x=6$	11	32	$\boxed{x=28}$
М			0	0	10	$\boxed{x=14}$
Σ	25		17		42	

$$2x + 2 = 14$$

$$x = \boxed{6}$$

$$|A| = 25$$

$$|Г| = 17$$

$$|Б| = 32$$

$$|Ж| = 28$$

$$|ЗЯ| = |ЗГ| + 2$$

$$|МГ| = 0$$

$$|МЗЯ| = 0$$

$$|U| = |A + Г| = 42$$

$$|БЖЯ| = ?$$

$$|З| = |U| - |Ж| = 42 - 28 = 14$$

$$|З| = |ЗГ| + |ЗЯ| = |ЗГ| + |ЗГ| + 2 \Rightarrow \boxed{|ЗГ| = 6}$$

$$\boxed{|ЗЯ| = 8}$$

$$\boxed{|ЖГ| = |Г| - |ЗГ| = 17 - 6 = 11}$$

$$\boxed{|ЖЯ| = |Я| - |ЗЯ| = 25 - 8 = 17}$$

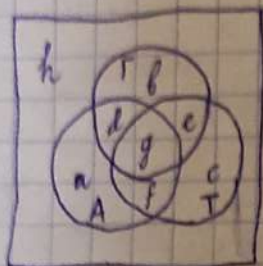
$$\boxed{|БЗГ| = |ЗГ| = 6}$$

$$\boxed{|БЖГ| = |ЖГ| = 11}$$

$$|БЗЯ| = |ЗЯ| = 8$$

$$|БЖЯ| = |Б| - |БЗЯ| - |БЗП| - |БЖП| = 32 - 8 - 6 - 11 = \boxed{7}$$

III.



$h = ?$

$d+e+f = ?$

$$\begin{cases} a+b+c+d+e+f+g+h=100 \\ a+d+f+g=80 \\ b+d+e+g=70 \\ c+e+f+g=60 \\ d+g=60 \\ f+g=50 \\ e+g=40 \\ g=30 \end{cases}$$

$$d = 30$$

$$f = 20$$

$$e = 10$$

$$a+b+c+2(d+e+f)+3g=210$$

$$a+b+c+d+e+f+g=90$$

$$h = 10$$

$$d+e+f = 60$$

III*.

	AX	AT	KX	KT	Σ
M		x	0	x+3	17
X	0	0	y	x	8
Σ					25

$$|A| = |K| + 1$$

$$|A| = 13 \quad |K| = 12$$

$$|KTM| = |KTX| + 3$$

$$|ATM| = |KTX|$$

$|KTX| - ?$

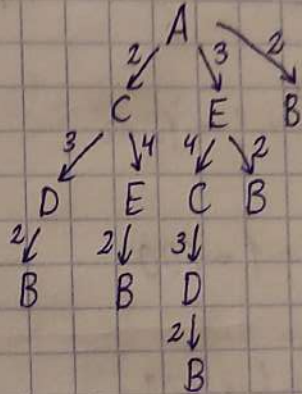
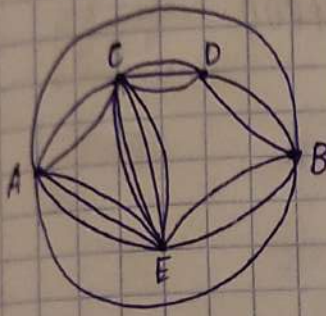
$$\begin{cases} x+y=8 \\ 2x+y+3=12 \end{cases}$$

$x=1$

$|KTX|=1$

II.

$\pi 5.$



$$\begin{aligned} &2 \cdot 3 \cdot 2 + 2 \cdot 4 \cdot 2 + \\ &+ 3 \cdot 4 \cdot 3 \cdot 2 + 3 \cdot 2 + 2 = \\ &= 12 + 16 + 72 + 6 + 2 = \\ &= 108 \end{aligned}$$

$\pi 6.$

$N = 10^4$

10 kpr. 10 kpr. 10 kpr. 10 kpr.
1-zureichen 2-zureichen 3-zureichen 4-zureichen

$\pi 7.$

$$\begin{cases} a_1 + a_2 + a_3 = 12 \\ a_1, a_2, a_3 \geq 1 \end{cases}$$

$\tilde{a}_1 = a_1 - 1, \tilde{a}_2 = a_2 - 1, \tilde{a}_3 = a_3 - 1$

$\tilde{a}_1 + \tilde{a}_2 + \tilde{a}_3 = 9$

$N = \sum_{i=0}^9 (9-i) = 1+2+\dots+9 = 45$ $N = \overline{C}_3^9 = C_{11}^9 = \frac{11!}{9! \cdot 2!} = 55$

$N = \overline{C}_9^3 = C_{11}^3 = \frac{11!}{3! \cdot 8!} = \frac{11 \cdot 10 \cdot 9}{2 \cdot 3} = 165$

+1

$|K|=12$

$|KTX|+3$

$|KTX|$

π8.

$$N = N_3 + N_5 + N_7 - 2N_{15} - 2N_{21} - 2N_{35} + 3N_{105}$$

$$1000 \leq n \leq 9999$$



$$N_3 = \frac{9999 - 1002}{3} + 1 = 3000$$

$$N_5 = \frac{9995 - 1000}{5} + 1 = 1800$$

$$N_7 = \frac{9996 - 1001}{7} + 1 = 1286$$

$$N_{15} = \frac{9990 - 1005}{15} + 1 = 600$$

$$N_{21} = \frac{9996 - 1008}{21} + 1 = 429$$

$$N_{35} = \frac{9975 - 1015}{35} + 1 = 257$$

$$N_{105} = \frac{9975 - 1050}{105} + 1 = 86$$

$$\begin{aligned} N &= 3000 + 1800 + 1286 - 2 \cdot 600 - 2 \cdot 429 - 2 \cdot 257 + 3 \cdot 86 = \\ &= 4800 + 86 - 2 \cdot 686 + 3 \cdot 86 = 4800 - 1200 + 2 \cdot 86 = \boxed{3772} \end{aligned}$$

π9.

$$N = N_{11,13} + N_{11,17} + N_{13,17} - 3N_{11,13,17} \quad 1000 \leq n \leq 2000$$

$$N_{11,13} = \frac{1867 - 1001}{143} + 1 = 63 \quad \frac{1859 - 1001}{143} + 1 = 7$$

$$N_{11,17} = \frac{1870 - 1122}{187} + 1 = 5$$

$$N_{\text{из}} = \frac{1989 - 1105}{221} + 1 = 5$$

$$N_{\text{ост}} = 0, \text{ так } 2431 > 2000.$$

$$N = 7 + 5 + 5 = 17$$

П10.

$$a) \begin{array}{cccc} a & b & c & d \\ \hline \text{H} & \text{H} & \text{H} & \text{H} \\ 0 & a & b & c \end{array}$$

$$N = 5 \cdot 5 \cdot 4 \cdot 3 = 300$$

$$b) \begin{array}{cccc} a & b & c & d \\ \hline g & & & \end{array}$$

$$N = 5 \cdot 6^3 = 1080$$

П11.

$100!$ - количество перестановок студентов на курсе.

$20!$ - количество перестановок студентов в группе

П.к. порядок студентов в группе не важен, но

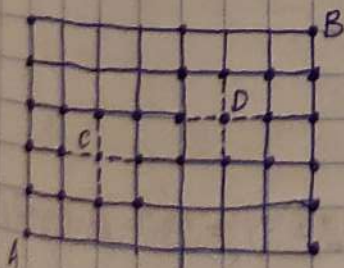
$$N = \frac{100!}{(20!)^5} = 109491541552511982098722568830981822088$$

$$3072063883928031640993360000$$

86 =

$$= 3772$$

П12.



$$N = N_{AB} - N_{AC} \cdot N_{CB} - N_{AD} \cdot N_{DB} + N_{AC} \cdot N_{CD} \cdot N_{DB}$$

$$N_{AB} = C_{12}^5 = \frac{12!}{5! \cdot 7!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{5 \cdot 4 \cdot 3 \cdot 2} = 792$$

$$N_{AC} = C_4^2 = \frac{4!}{2! \cdot 2!} = 6$$

$$N_{CB} = C_8^3 = \frac{8!}{3! \cdot 5!} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2} = 56$$

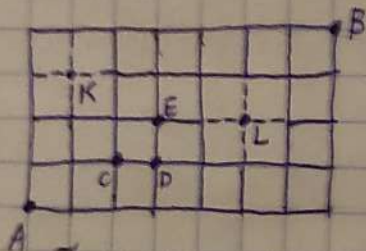
$$N_{AD} = C_8^3 = 56$$

$$N_{DB} = C_4^2 = 6$$

$$N_{AB} = C_4^1 = 4$$

$$N = 792 - 6 \cdot 56 - 56 \cdot 6 + 6 \cdot 4 \cdot 6 = 12(66 - 56) + 36 \cdot 4 = 120 + 144 = \boxed{264}$$

11.13.



$$p = \frac{N_{AC} \cdot \tilde{N}_{EB}}{\tilde{N}_{AB}}$$

$$N_{AC} = C_3^1 = 3$$

$$\tilde{N}_{EB} = N_{EB} - N_{EL} \cdot N_{LB}$$

$$N_{EB} = C_6^2 = \frac{6!}{2! \cdot 4!} = \frac{6 \cdot 5}{2} = 15$$

$$N_{EL} = C_2^0 = 1$$

$$N_{LB} = C_4^2 = 6$$

$$\tilde{N}_{EB} = 15 - 1 \cdot 6 = 9$$

$$\tilde{N}_{AB} = N_{AB} - N_{AK} \cdot N_{KB} - N_{AL} \cdot N_{LB}$$

N_{KL} is given above, the number of ways to choose K, L

$$N_{AB} = C_{11}^4 = \frac{11!}{4! \cdot 7!} = \frac{11 \cdot 10 \cdot 9 \cdot 8}{4 \cdot 3 \cdot 2} = 330$$

$$N_{AK} = C_4^1 = 4$$

$$N_{KB} = C_7^1 = 7$$

$$N_{AL} = C_7^2 = \frac{7!}{2! \cdot 5!} = \frac{7 \cdot 6}{2} = 21$$

$$\tilde{N}_{AB} = 330 - 4 \cdot 7 - 21 \cdot 6 = 330 - 28 - 126 = 176$$

$$p = \frac{3 \cdot 9}{176} = \frac{27}{176} \approx \boxed{0,1534} = 15,34\%$$

π14.

$$a) p = \frac{\bar{N}}{N}$$

$$N = 10!$$

$$\bar{N} = 10! - 1 \cdot 9!$$

$$p = \frac{10! - 9!}{10!} = 1 - \frac{1}{10} = \boxed{0,9} = 90\%$$

$$b) p = \frac{\bar{N}}{N}$$

$$N = 10!$$

$$\bar{N} = 10! - 5! \cdot \frac{10!}{5! \cdot 5!} \cdot \bar{N}_5$$

$$\bar{N}_5 = \bar{N}_5 = 10! / 5! - 4!$$

$$\bar{N} = \frac{10! (5! - 4!)}{5! \cdot 5!} = \frac{10!}{5!} \left(1 - \frac{1}{5}\right) = \frac{10! - 2 \cdot 9!}{5!}$$

$$p = \frac{\bar{N}}{N} = \frac{10!}{10! \cdot 5!} - \frac{2 \cdot 9!}{5! \cdot 10!} = \frac{1}{120} - \frac{2}{1200} = \frac{1}{150} = \boxed{0,0067} = 0,67\%$$

π15.

$$x_1 + x_2 + x_3 + x_4 + x_5 \geq 37$$

$$0 \leq x_1, \dots, x_5 \leq 9, x_1, \dots, x_5 \in \mathbb{Z}$$

$$\bar{x}_i = 9 - x_i \quad \forall i \quad \bar{x}_i \in [0; 9] \cap \mathbb{Z}$$

$$9 \cdot 5 - \bar{x}_1 - \bar{x}_2 - \bar{x}_3 - \bar{x}_4 - \bar{x}_5 \geq 37$$

$$\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 + \bar{x}_5 \leq 8$$

$$\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 + \bar{x}_5 + \bar{x}_6 = 8$$

$$N = \bar{C}_6^8 = C_{13}^8 = \frac{13!}{8! \cdot 5!} = \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9}{8 \cdot 4 \cdot 3 \cdot 2} = 13 \cdot 11 \cdot 9 = \boxed{1287}$$

III.

117.

1) $f(x, y, z) = (x+y)(\bar{x}+z) + y(y+\bar{z})$

a)

x	y	z	$(x+y)(\bar{x}+z)$	$y(y+\bar{z})$	f
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	0	0
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	0	0	0

$$f(x, y, z) = x\bar{y}\bar{z} + xy\bar{z}$$

$$f(x, y, z) = (x+y+z)(x+y+\bar{z})(x+\bar{y}+z)(x+\bar{y}+\bar{z})(\bar{x}+y+\bar{z})(\bar{x}+\bar{y}+\bar{z})$$

d) $f = (x+y)(\bar{x}+z) + y(y+\bar{z}) = (x+y)x\bar{z} + y\bar{y}z = 0$

$$= x\bar{x}\bar{z} + xy\bar{z} = x\bar{z}(y+1) = x\bar{z} = x\bar{z}(y+\bar{y}) = \boxed{xy\bar{z} + x\bar{y}\bar{z}}$$

$$f = x\bar{z} = x(\bar{y}z + y\bar{z})(\bar{x} + yz + \bar{y}\bar{z})(\bar{z} + xy + \bar{x}\bar{y})$$

$$= x\bar{z} + y\bar{y} = (x\bar{z} + y)(x\bar{z} + \bar{y}) = (x+y)(\bar{z} + y)(x+\bar{y})(\bar{z} + \bar{y}) =$$

$$= (x+y+z)(x+y+\bar{z})(\bar{z}+y+x)(\bar{z}+y+\bar{x})(x+\bar{y}+z)(x+\bar{y}+\bar{z})(\bar{z}+\bar{y}+x)(\bar{z}+\bar{y}+\bar{x})$$

$$= \boxed{(x+y+z)(x+y+\bar{z})(\bar{x}+y+\bar{z})(x+\bar{y}+z)(x+\bar{y}+\bar{z})(\bar{x}+\bar{y}+\bar{z})}$$

e) $g(x, y, z) = (\bar{x}+\bar{y}+z)(x+yz)$

a)

x	y	z	$\bar{x}+\bar{y}+z$	$x+yz$	g
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	1	0	1
0	1	1	1	1	0
1	0	0	0	1	0
1	0	1	1	1	0
1	1	0	0	1	1
1	1	1	1	1	0

$$g(x, y, z) = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + xy\bar{z}$$

$$g(x, y, z) = (x + \bar{y} + \bar{z})(\bar{x} + y + z)(\bar{x} + y + \bar{z})(\bar{x} + \bar{y} + z)$$

$$\begin{aligned} f) g &= (\bar{x} + \bar{y} + z)(x + yz) = (\bar{x} + \bar{y} + z) + (\bar{x} + \bar{y} + z) = \\ &= ((\bar{x} + \bar{y})\bar{z}) + \bar{x}(yz) = xy\bar{z} + \bar{x}(\bar{y} + \bar{z}) = xy\bar{z} + \bar{x}\bar{y} + \bar{x}\bar{z} = \\ &= xy\bar{z} + \bar{x}\bar{y}(z + \bar{z}) + \bar{x}\bar{z}(y + \bar{y}) = xy\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}\bar{y}\bar{z} = \\ &= \boxed{xy\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z}} \end{aligned}$$

$$g = xy\bar{z} + \bar{x}(\bar{y} + \bar{z}) = (xy\bar{z} + \bar{x})(xy\bar{z} + \bar{y} + \bar{z}) = (\bar{x} + xy)(\bar{x} + \bar{z})$$

$$(\bar{x} + \bar{y} + \bar{z})(y\bar{z} + \bar{y} + \bar{z}) = (\bar{x} + x)(\bar{x} + y)(\bar{x} + \bar{z})(x + \bar{y} + \bar{z})(\bar{y} + \bar{z} + y)(\bar{y} + \bar{z} + \bar{z}) =$$

$$= (\bar{x} + \bar{y} + \bar{z})(\bar{x} + y)(\bar{x} + \bar{z})(\bar{y} + \bar{z}) = (x + \bar{y} + \bar{z})(\bar{x} + y + z)(\bar{x} + y + \bar{z})(\bar{x} + y + \bar{z})$$

$$(\bar{x} + \bar{y} + \bar{z}) + (x + \bar{y} + \bar{z})(\bar{x} + \bar{y} + \bar{z}) = \boxed{(x + \bar{y} + \bar{z})(\bar{x} + y + z)(\bar{x} + y + \bar{z})(\bar{x} + \bar{y} + \bar{z})}$$

$$3) h(x, y, z) = (x \oplus \bar{y})(\bar{x}z \oplus y)(x + y + \bar{z})$$

x	y	z	$x \oplus \bar{y}$	$\bar{x}z \oplus y$	$x + y + \bar{z}$	h
0	0	0	1	0	1	0
0	0	1	1	1	0	0
0	1	0	0	1	1	0
0	1	1	0	0	1	0
1	0	0	0	0	1	0
1	0	1	0	0	1	0
1	1	0	1	1	1	1
1	1	1	1	1	1	1

$$h(x, y, z) = xy\bar{z} + xyz$$

$$h(x, y, z) = (x + y + z)(x + y + \bar{z})(x + \bar{y} + z)(x + \bar{y} + \bar{z})(\bar{x} + y + z)(\bar{x} + y + \bar{z})$$

$$\begin{aligned} f) h &= xy(x \oplus \bar{y})(\bar{x}z \oplus y)(x + y + \bar{z}) = (\bar{x}\bar{y} + xy)(\bar{x}z\bar{y} + \bar{x}z\bar{y})(x + y + \bar{z}) = \\ &= (\bar{x}\bar{y} + xy)(xy + \bar{x}y + \bar{x}z\bar{y})(x + y + \bar{z}) = (xy + \bar{x}y\bar{z})(xy + \bar{x}y + \bar{x}z\bar{y}) = \end{aligned}$$

$$= xy + xy\bar{z} + xy(\bar{z} + 1) = xy = xy(z + \bar{z}) = \boxed{xyz + xy\bar{z}}$$

$$h = (x \oplus \bar{y})(\bar{x}z \oplus y)(x + y + \bar{z}) = (x + \bar{y})(\bar{x} + y)(\bar{x}z + y)(x + \bar{z} + \bar{y})(x + y + \bar{z}) =$$

$$= (x + \bar{y})(\bar{x} + y)(\bar{x} + y)(z + y)(x + \bar{z} + \bar{y})(x + y + \bar{z}) = (x + \bar{y} + z)(x + \bar{y} + \bar{z}) \cdot$$

$$\cdot (\bar{x} + y + z)(\bar{x} + y + \bar{z})(x + y + z)(\bar{x} + y + z)(x + \bar{y} + \bar{z})(x + y + \bar{z}) =$$

$$= \boxed{(x + \bar{y} + z)(x + \bar{y} + \bar{z})(\bar{x} + y + z)(\bar{x} + y + \bar{z})(x + y + z)(x + y + \bar{z})}$$

$$4) k(x, y, z) = (xy + z)((x \oplus y) + \bar{z})$$

a)	x	y	z	xy+z	(x⊕y)+z̄	k
	0	0	0	0	1	0
	0	0	1	1	0	1
	0	1	0	0	1	0
	0	1	1	1	1	0
	1	0	0	0	1	0
	1	0	1	1	1	0
	1	1	0	1	1	0
	1	1	1	1	0	1

$$k(x, y, z) = \bar{x}\bar{y}z + xyz$$

$$k(x, y, z) = (x + y + z)(x + \bar{y} + \bar{z})(x + \bar{y} + \bar{z})(\bar{x} + y + z)(\bar{x} + y + \bar{z})(\bar{x} + \bar{y} + z)$$

$$b) k = (xy + z)((x \oplus y) + \bar{z}) = (xy + z)(x \oplus y)z = (xy + z)(\bar{x} + y)(x + \bar{y})z =$$

$$= (zxy + z)(xy + \bar{x}\bar{y}) = z(xy + \bar{x}\bar{y}) = \boxed{xyz + \bar{x}\bar{y}z}$$

$$k = z(xy + z)(\bar{x} + y)(x + \bar{y}) = z(\bar{x} + y)(x + \bar{y}) = (z + x)(z + \bar{x})(\bar{x} + y)(x + \bar{y}) =$$

$$= (z + x + y)(z + x + \bar{y})(z + \bar{x} + y)(z + \bar{x} + \bar{y})(\bar{x} + y + z)(\bar{x} + y + \bar{z})(x + \bar{y} + z)(x + \bar{y} + \bar{z}) =$$

$$= \boxed{(x + y + z)(x + \bar{y} + z)(\bar{x} + y + z)(\bar{x} + \bar{y} + z)(\bar{x} + y + \bar{z})(x + \bar{y} + \bar{z})}$$

III 18.

$$f(x, y, z) = \{10110011\}, g(x, y, z) = \{10011001\}, h(x, y, z) = \{11010101\}.$$

$$(y + \bar{z}) = \bar{z}$$

1)

x	y	z	f
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$$\begin{array}{l}
 a_0 = 1 \\
 a_x = 1 \\
 a_y = 0 \\
 a_{yz} = 1 \\
 a_{xz} = 1 \\
 a_{xy} = 1 \\
 a_{xyz} = 1
 \end{array}$$

$$f(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}yz + xy\bar{z} + xyz$$

$$f(x,y,z) = (x+y+\bar{z})(\bar{x}+y+z)(\bar{x}+y+\bar{z})$$

$$f(x,y,z) = 1 \oplus x \oplus z \oplus xy \oplus xz \oplus yz \oplus xyz$$

2)

x	y	z	g
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$\begin{array}{l}
 a_0 = 1 \\
 a_x = 1 \\
 a_y = 1 \\
 a_{yz} = 1 \\
 a_{xz} = 1 \\
 a_{xy} = 1 \\
 a_{xyz} = 1
 \end{array}$$

$$g(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}yz + x\bar{y}\bar{z} + xyz$$

$$g(x,y,z) = (x+y+\bar{z})(x+\bar{y}+z)(\bar{x}+y+\bar{z})(\bar{x}+\bar{y}+z)$$

$$g(x,y,z) = 1 \oplus y \oplus z$$

$$h(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}yz + \bar{x}yz + \bar{x}yz + xyz$$

$$h(x,y,z) = (x+\bar{y}+z)(\bar{x}+y+z)(\bar{x}+\bar{y}+z)$$

$$h(x,y,z) = 1 \oplus x \oplus y \oplus xy \oplus xz \oplus yz \oplus xyz$$

3)

x	y	z	h
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$\begin{array}{l}
 a_0 = 1 \\
 a_x = 0 \\
 a_y = 1 \\
 a_{yz} = 1 \\
 a_{xz} = 1 \\
 a_{xy} = 1 \\
 a_{xyz} = 1
 \end{array}$$

	f	g	h
T_0	-	-	-
T_1	+	+	+
S	-	-	-
L	-	+	-
M	-	-	-

$\Rightarrow$ система не является полной.

П19.

$$f_1(x, y, z) = (x + y)(\bar{x} + \bar{y}) = x \oplus y$$

$$f_2(x, y, z) = xy \oplus z$$

$$f_3(x, y, z) = xy \sim z = \bar{f}_2(x, y, z)$$

$$f_4(x, y, z) = xy + xz + yz$$

x	y	z	f_1	f_2	f_3	f_4
0	0	0	0	0	1	0
0	0	1	0	1	0	0
0	1	0	1	0	1	0
0	1	1	1	1	0	1
1	0	0	1	0	1	0
1	0	1	1	1	0	1
1	1	0	0	1	0	1
1	1	1	0	0	1	1

	f_1	f_2	f_3	f_4
--	-------	-------	-------	-------

T_0	+	+	-	+
-------	---	---	---	---

T_1	-	-	+	+
-------	---	---	---	---

S	-	-	-	+
-----	---	---	---	---

L	+	-	-	-
-----	---	---	---	---

M	-	-	-	+
-----	---	---	---	---

$\Rightarrow$ система является полной.

П20.

$$f(x_1, x_2, x_3, x_4) = x_1 + (x_2 \oplus x_3) + \bar{x}_4$$

T_0	T_1	S	L	M
-	+	-	-	-

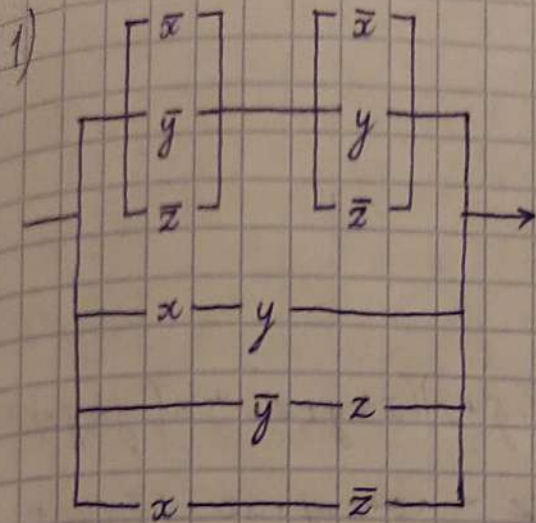
$$f = x_1(x_2 \oplus x_3) \oplus x_1 \oplus x_2 \oplus x_3 + \bar{x}_4 =$$

$$= (x_1 x_2 \oplus x_1 x_3 \oplus x_1 \oplus x_2 \oplus x_3)(x_4 \oplus 1) \oplus$$

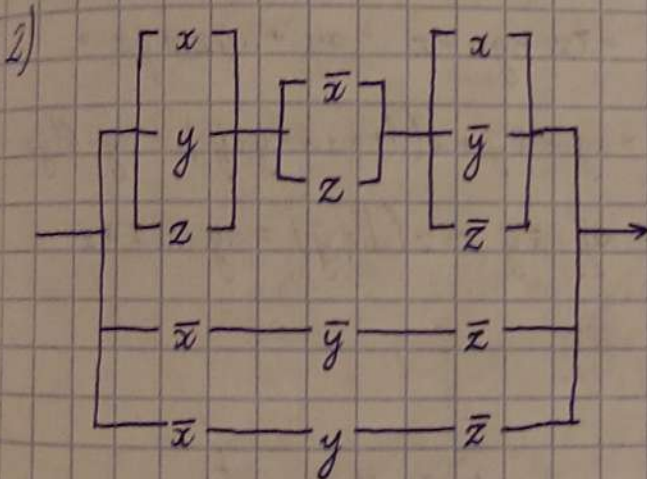
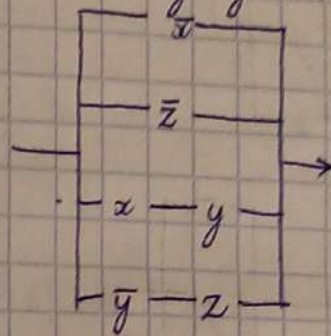
$$\oplus x_1 x_2 \oplus x_1 x_3 \oplus x_1 \oplus x_2 \oplus x_3 \oplus x_4 \oplus 1 \Rightarrow f \notin L$$

x_1	x_2	x_3	x_4	f
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

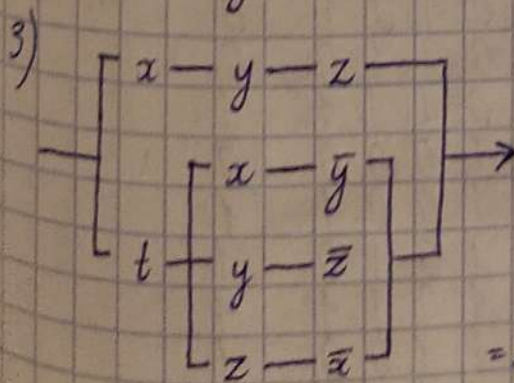
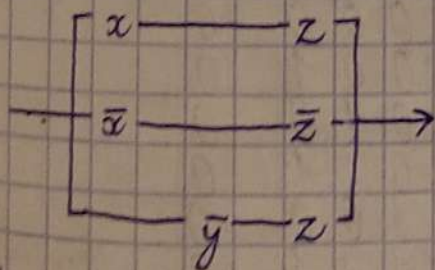
№ 21.



$$f = (\bar{x} + \bar{y} + \bar{z})(x + y + z) + xyz + \bar{x}\bar{y}\bar{z} = \bar{x} + \bar{z} + xy + \bar{y}z + x\bar{z} = \bar{x} + \bar{z} + xy + \bar{y}z$$



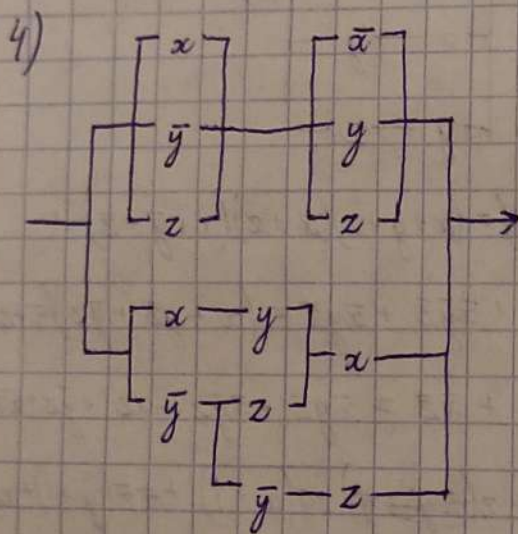
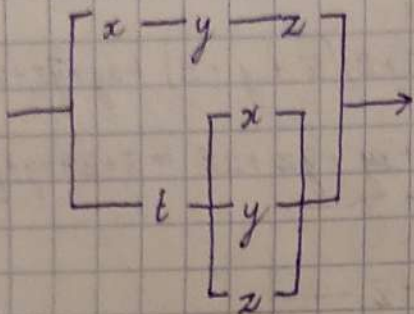
$$f = (x + y + z)(\bar{x} + z)(x + \bar{y} + \bar{z}) + \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} = (x + y\bar{z} + \bar{y}z)(\bar{x} + z) + \bar{x}\bar{z} = \bar{x}y\bar{z} + \bar{x}\bar{y}z + xz + \bar{y}z + \bar{x}\bar{z} = z(\bar{x}\bar{y} + \bar{y}) + \bar{x}\bar{z}(y + 1) + xz = \bar{y}z + \bar{x}\bar{z} + xz$$



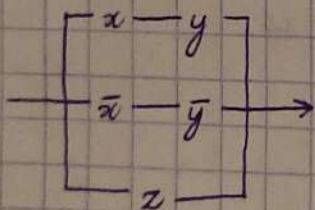
$$f = xyz + t(\bar{x}\bar{y} + \bar{y}\bar{z} + \bar{z}\bar{x}) = xyz(1 + t) + t(\bar{x}\bar{y} + \bar{y}\bar{z} + \bar{z}\bar{x}) = xyz + t(xyz + \bar{x}\bar{y} + \bar{x}\bar{y}z + \bar{y}\bar{z} + \bar{x}yz + \bar{z}\bar{x}) = xyz + t(x(\bar{y}z + \bar{y}) + y(\bar{x}z + \bar{z}) + z(\bar{x}\bar{y})) = xyz + t(x(\bar{y} + z) + y(\bar{z} + \bar{x}) + z(\bar{x} + \bar{y})) = xyz + t(x\bar{y} + xz + y\bar{z} + y\bar{x} + z\bar{x} + zy) =$$

f
1
0
1
1
1
1
1
0
1
1
1
1
1
1
1
1

$$= xyz + t(x(y+\bar{y}) + y(z+\bar{z}) + z(x+\bar{x})) = xyz + t(x+y+z)$$

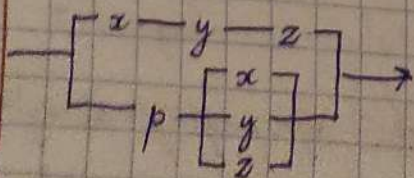


$$\begin{aligned} f &= (x+\bar{y}+z)(\bar{x}+y+z) + x(\bar{y}z) + \\ &+ \bar{y}\bar{y}z = \cancel{xz} + \bar{x}y + xz + \bar{x}\bar{y} + \bar{y}z + \\ &+ \bar{x}z + yz + z + \bar{x}y + \bar{x}\bar{y}z + \bar{y}z = \\ &= \bar{x}y + \bar{x}\bar{y} + z(1+\bar{x}+y+\bar{y}) + \bar{x}y + \bar{y}z \\ &= \bar{x}y + \bar{x}\bar{y} + z(1+\bar{y}) = \bar{x}y + \bar{x}\bar{y} + z \end{aligned}$$



M22.

$$\begin{aligned} p, x, z \quad f &= p(xy+yz+zx+x+y+z) + \\ &+ xyz = p(x(y+1)+y(z+1)+z(x+1))+xyz = \\ &= p(x+y+z) + xyz \end{aligned}$$



p	x	y	z	f
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

M 23.

$$s = s_0 \oplus f$$

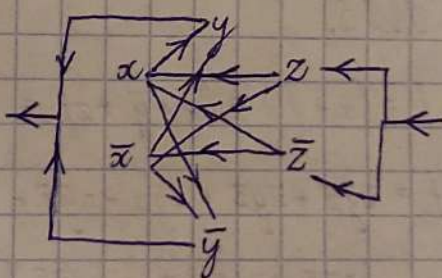
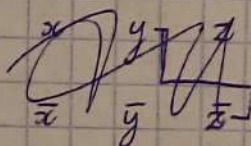
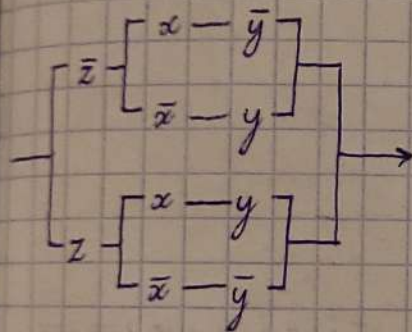
$$s = s_0 \oplus x \oplus y \oplus z$$

$$f = (xy + \bar{x}y) \oplus z =$$

$$= (xy + \bar{x}y)\bar{z} + (xy + \bar{x}y)z =$$

$$= (xy + \bar{x}y)\bar{z} + (\bar{x} + y)(x + \bar{y})z =$$

$$= x\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}y\bar{z} + x\bar{y}z = \bar{z}(xy + \bar{x}y) + z(xy + \bar{x}\bar{y})$$



M 24.

x	y	z	f*	f
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

$a_0 = 1$	1	1	0	0	1	1	1
$a_z = 0$	0	0	1	0	1	0	0
$a_y = 0$	0	1	1	1	1	0	0
$a_{yz} = 1$	0	0	0	0	1	0	0
$a_x = 1$	0	0	0	1	0	0	1
$a_{xz} = 1$	0	0	1	0	1	0	0
$a_{xy} = 1$	1	1	0	0	0	0	0
$a_{xyz} = 0$	0	0	0	0	0	0	0

$$f = 1 \oplus x \oplus xy \oplus xz \oplus yz$$

$$ff = (x + \bar{y} + \bar{z})(\bar{x} + y + z)$$

I HOPE

$$f = (x + \bar{y} + \bar{z})(\bar{x} + y + z) = xy + xz + \bar{x}\bar{y} + yz + \bar{x}\bar{z} + y\bar{z} - \text{comp. DNF.}$$

$$f = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}y\bar{z} + x\bar{y}z + xy\bar{z} + xyz$$

	$\bar{x}\bar{y}\bar{z}$	$\bar{x}y\bar{z}$	$\bar{x}y\bar{z}$	$x\bar{y}z$	$xy\bar{z}$	xyz
xy					+	+
$\bar{x}z$				+		+
$\bar{x}\bar{y}$	+	+				
$y\bar{z}$		+		+		
$\bar{x}\bar{z}$	+		+			
$y\bar{z}$			+		+	

Исходящие не являются частью минимальных выражений.

$$f = xy + \bar{y}z + \bar{x}\bar{z}$$

$$\bar{f} = xz + \bar{x}\bar{y} + y\bar{z}$$

М25.

x	y	z	f
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

$$f = (x + y + \bar{z})(\bar{x} + \bar{y} + z) = x\bar{y} + xz + \bar{x}\bar{y} + yz + \bar{x}\bar{z} + y\bar{z}$$

$$f = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}yz + x\bar{y}\bar{z} + x\bar{y}z + xyz + xyz = D_2(f)$$

	$\bar{x}\bar{y}\bar{z}$	$\bar{x}y\bar{z}$	$\bar{x}yz$	$x\bar{y}\bar{z}$	$x\bar{y}z$	xyz
$x\bar{y}$				+	+	
xz					+	+
$\bar{x}\bar{y}$		+	+			
$y\bar{z}$			+			+
$\bar{x}\bar{z}$	+	+				
$\bar{y}\bar{z}$	+			+		

Минимальные: $xz + yz + \bar{x}\bar{z} + \bar{y}\bar{z}$; $xz + \bar{x}\bar{y} + yz + \bar{x}\bar{z}$

$x\bar{y} + \bar{x}\bar{y} + yz + \bar{y}\bar{z}$; $x\bar{y} + xz + \bar{x}\bar{y} + \bar{x}\bar{z}$.

Максимальные: $x\bar{y} + yz + \bar{x}\bar{z}$; $xz + \bar{x}\bar{y} + \bar{y}\bar{z}$.

IV.

M26.

$$1) f(x, y, z) = (y \sim (\overline{x+z})) (y \rightarrow xz) \sim x\bar{y}\bar{z}$$

a)

x	y	z	$y \sim (\overline{x+z})$	$y \rightarrow xz$	$x\bar{y}\bar{z}$	f
0	0	0	0	0	0	1
0	0	1	1	1	0	0
0	1	0	1	0	0	1
0	1	1	0	0	0	1
1	0	0	1	1	1	1
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	1	0	1	0	1

$$f = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}yz + x\bar{y}\bar{z} + xy\bar{z} + xyz$$

$$f = (x + y + \bar{z})(\bar{x} + y + \bar{z})$$

$$\begin{aligned} d) f &= (y \sim (\overline{x+z})) (y \rightarrow xz) \sim x\bar{y}\bar{z} = (y \sim \bar{x}\bar{z})(y \rightarrow xz) \sim x\bar{y}\bar{z} = \\ &= (y\bar{x}\bar{z} + \bar{y}(x+z))(\bar{y} + xz) \sim x\bar{y}\bar{z} = (\bar{y}(x+z) + yxz(x+z)) \sim x\bar{y}\bar{z} = \\ &= (\bar{y}x + \bar{y}z)(1 + xz) \sim x\bar{y}\bar{z} = (\bar{y}x + \bar{y}z) \sim x\bar{y}\bar{z} = x\bar{y}\bar{z}(\bar{y}x + \bar{y}z) + \\ &+ (\bar{x}\bar{y}\bar{z})(\bar{y}x + \bar{y}z) = x\bar{y}\bar{z} + (\bar{x} + y + z)(y + \bar{x})(y + \bar{z}) = x\bar{y}\bar{z} + (\bar{x} + y + z)(y + \bar{x}\bar{z}) = \end{aligned}$$

$$\begin{aligned} &= x\bar{y}\bar{z} + \bar{x}y + y + zy + \bar{x}\bar{z} + \bar{x}y\bar{z} = x\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}yz + \bar{x}y\bar{z} + xyz + xy\bar{z} + \\ &+ \bar{x}yz + \bar{x}y\bar{z} + xyz + \bar{x}yz + \bar{x}y\bar{z} + \bar{x}\bar{y}\bar{z} = \boxed{x\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}yz + xyz + xy\bar{z} + \bar{x}\bar{y}\bar{z}} \end{aligned}$$

$$f = x\bar{y}\bar{z} + \bar{x}y + y + zy + \bar{x}\bar{z} + \bar{x}y\bar{z} = y(1 + \bar{x} + \bar{z}) + \bar{z}(x\bar{y} + \bar{x} + \bar{z}y) =$$

$$= y + \bar{z}(x\bar{y} + \bar{x}) = y + \bar{z}(\bar{x} + x)(\bar{x} + y) = y(1 + \bar{z}) + \bar{z}(\bar{x} + y) =$$

$$= y + \bar{z}(\bar{x} + y + y) = y + \bar{z} = \boxed{(x + y + \bar{z})(\bar{x} + y + \bar{z})}$$

$$2) g(x, y, z) = z(x \rightarrow y) + (\bar{z} \rightarrow \bar{x})$$

a)

x	y	z	$z(x \rightarrow y)$	$\bar{z} \rightarrow \bar{x}$	g
0	0	0	0	1	1
0	0	1	1	1	1
0	1	0	0	1	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	0	0	0
1	1	1	1	1	1

$g = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + x\bar{y}z + xyz$
 $g = (\bar{x} + y + z)(\bar{x} + \bar{y} + z)$

$$b) g = z(x \rightarrow y) + (\bar{z} \rightarrow \bar{x}) = z(\bar{x} + y) + (z + \bar{x}) = z\bar{x} + zy + z + \bar{x} =$$

$$= z + \bar{x} = xz + \bar{x}z + \bar{x}\bar{z} = \boxed{xyz + x\bar{y}z + \bar{x}yz + \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}\bar{y}z}$$

$$g = z + \bar{x} = \boxed{(\bar{x} + y + z)(\bar{x} + \bar{y} + z)}$$

$$3) h(x, y, z) = (x \sim (\bar{y} \rightarrow (z + xy))) \rightarrow (xy + \bar{z})$$

a)

x	y	z	$(z + xy)$	$x \sim (\bar{y} \rightarrow (z + xy))$	$xy + \bar{z}$	h
0	0	0	1	0	1	1
0	0	1	0	1	0	0
0	1	0	1	0	1	1
0	1	1	0	0	0	1
1	0	0	1	1	1	1
1	0	1	0	0	0	1
1	1	0	0	1	1	1
1	1	1	0	1	1	1

$$h = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + x\bar{y}z + xy\bar{z} + xyz$$

$$h = x + y + \bar{z}$$

$$b) h = (x \sim (\bar{y} \rightarrow (z + xy))) \rightarrow (xy + \bar{z}) = (x \sim (y + \bar{z}(\bar{x} + \bar{y}))) \rightarrow (xy + \bar{z}) =$$

$$= (x \sim (y + \bar{x}\bar{z} + \bar{y}\bar{z})) \rightarrow (xy + \bar{z}) = (x \sim (y + \bar{z})) \rightarrow (xy + \bar{z}) =$$

$$= (x(y + \bar{z}) + \bar{x}(y + \bar{z})) \rightarrow (xy + \bar{z}) = (xy + x\bar{z} + \bar{x}\bar{y}z) \rightarrow (xy + \bar{z}) =$$

$$\begin{aligned}
 &= (\bar{x}y + x\bar{z} + \bar{x}y\bar{z}) + xy + \bar{z} = (\bar{x} + y)(\bar{x} + z)(x + y + \bar{z}) + xy + \bar{z} = \\
 &= (\bar{x} + y\bar{z})(x + y + \bar{z}) + xy + \bar{z} = \bar{x}y + \bar{x}\bar{z} + \bar{x}y\bar{z} + xy + \bar{z} = \\
 &= \bar{x}yz + \bar{x}y\bar{z} + \bar{x}y\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz + xy\bar{z} + xy\bar{z} + \bar{x}y\bar{z} + \bar{x}y\bar{z} + \bar{x}y\bar{z} = \\
 &= \boxed{\bar{x}yz + \bar{x}y\bar{z} + \bar{x}y\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz + xy\bar{z} + xy\bar{z}}
 \end{aligned}$$

$$\begin{aligned}
 h &= \bar{x}y + \bar{x}\bar{z} + \bar{x}y\bar{z} + xy + \bar{z} = y(\bar{x} + \bar{z}) + \bar{z}(1 + \bar{x}) + x\bar{y}z = \\
 &= y + \bar{z} + x\bar{y}z = y(1 + xz) + \bar{z}(1 + x\bar{y}) + x\bar{y}z = \\
 &= y + \bar{z} + x(yz + \bar{y}\bar{z} + y\bar{z} + \bar{y}z) = \boxed{x + y + \bar{z}}
 \end{aligned}$$

$$4) k(x, y, z) = ((x \rightarrow \bar{y})(y \rightarrow \bar{z})) \rightarrow (z \rightarrow \bar{x})$$

x	y	z	$x\bar{y}$	$y\bar{z}$	$(x\bar{y})(y\bar{z})$	$(z \rightarrow \bar{x})$	k	$k = \bar{x}yz + \bar{x}y\bar{z} + \bar{x}y\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z} + xyz$
0	0	0	0	1	0	1	1	
0	0	1	1	1	1	1	1	$k = \bar{x} + y + \bar{z}$
0	1	0	1	1	1	1	1	
0	1	1	1	0	0	1	1	
1	0	0	1	1	1	1	1	
1	0	1	1	1	1	0	0	
1	1	0	0	1	0	1	1	
1	1	1	0	0	0	0	1	

$$\begin{aligned}
 i) k &= (x \rightarrow \bar{y})(y \rightarrow \bar{z}) \rightarrow (z \rightarrow \bar{x}) = (\bar{x} + y)(\bar{y} + \bar{z}) \rightarrow (\bar{z} + \bar{x}) = \\
 &= (\bar{y} + \bar{x}\bar{z}) + (\bar{x} + \bar{z}) = y(x + z) + \bar{x} + \bar{z} = xy + yz + \bar{x} + \bar{z} = \\
 &= xyz + xy\bar{z} + x\bar{y}z + \bar{x}yz + \bar{x}yz + \bar{x}yz + \bar{x}yz + \bar{x}yz + xy\bar{z} + x\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}y\bar{z} = \\
 &= \boxed{xyz + xy\bar{z} + \bar{x}yz + \bar{x}yz + \bar{x}yz + \bar{x}yz + \bar{x}yz + \bar{x}yz} \\
 k &= xy + yz + \bar{x} + \bar{z} = xy + yz + \bar{x}(1 + y) + \bar{z}(1 + y) = y(x + z + \bar{x} + \bar{z}) + \bar{x} + \bar{z} = \\
 &= \boxed{\bar{x} + y + \bar{z}}
 \end{aligned}$$

М27.

$$\begin{cases} A \rightarrow \bar{B} = 1 & \bar{A} + \bar{B} = 1 \\ D \rightarrow A = 1 & \bar{D} + A = 1 \\ \bar{B} \rightarrow (C + \bar{A}) = 1 & B + C + \bar{A} = 1 \\ \bar{D} \rightarrow A\bar{C} = 1 & D + A\bar{C} = 1 \end{cases}$$

$$(D + A\bar{C})(\bar{D} + A) = 1$$

$$DA + \bar{D}A\bar{C} + A\bar{C} = 1$$

$$A(D + \bar{C} + \bar{C}\bar{D}) = 1$$

$$A = 1$$

$$1 \cdot 0 + \bar{B} = 1, \Rightarrow B = 0$$

$$0 + C + 0 = 1 \Rightarrow C = 1$$

$$D + 1 \cdot 0 = 1, \Rightarrow D = 1$$

A, C и D сданы, B не сдан

М28.

Предположим, что варенье съел Вася. Тогда первое утверждение Васи, "Вова съел" - ложь, а второе, "Аня не съ" - правда, что противоречит условию; значит, Вася не съел варенье.

Предположим, что варенье съел Аня. Тогда первое утверждение Ани, "Вова не съ" - правда, а второе - "Это Вася" - ложь, что противоречит условию; значит, Аня не съел варенье.

Значит, верные все Вова. Убеждаемся, что при этом не возникает противоречия: Анна в обоих играх сыграла, а Вова в обеих играх сказал правду. Вова сказал правду в первой игре и, стало быть, сказал правду и во второй игре.

Вова.

№29.

Предположим, что С говорит правду, тогда А и В лгут. Но тогда А говорит правду, т.к. он говорит, что В лжет. Противоречие.

Значит, С лжет. Тогда В говорит правду, т.к. он говорит, что С лжет. Тогда А лжет, так как он говорит, что В лжет. В говорит правду, А и С лгут.